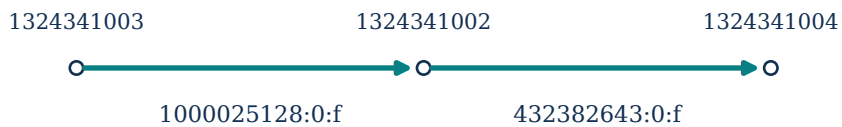


# Local construction of the time-expanded graph

a Directed physical road chain



b Object mapping

Physical node  $\rightarrow$  time-indexed states  
Road traversal  $\rightarrow$  later-time movement  
Waiting preserves the physical node  
No separate turn-connector layer

c Exact saved movement and waiting arcs

